

Endogeneity

ECON20222 - Lecture 4

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Recap

- Recall the simple regression model

$$y = \alpha + \beta x + u \quad (1)$$

- **The exogeneity assumption**

$$E(u \mid x) = E(ux) = Cov(u, x) = 0 \quad (A2)$$

OLS by construction imposes the sample analogue:

$$\sum_i \hat{u}_i x_i = 0$$

Is A2 true? What happens if A2 holds, and what happens if it does not?

Exogeneity

- $Cov(u, x) = 0$ is **true**
 - ① x is said to be *exogenous* (this is an econometrics definition, different to that used by economists);
 - ② the OLS estimator $\hat{\beta}$ is said to be **unbiased**¹; and
 - ③ the effect of x on y (given by the OLS estimate $\hat{\beta}$) is said to be **causal**.

¹A related property is *consistency*, which is unbiasedness, but only in “large” samples.

Endogeneity

- $Cov(u, x) = 0$ is **not true**
 - ① x is said to be **endogenous** (again, this is an econometrics definition);
 - ② the OLS estimator $\hat{\beta}$ is said to be **biased**; and
 - ③ the effect of x on y cannot be said to be **causal**. Rather, it is just a correlation.

In this lecture we explore three ways in which this can happen.

- ▶ Omitted Variables Bias
- ▶ Measurement Error Bias
- ▶ Simultaneity Bias

Omitted Variables Bias (OVB)

- Suppose the **true model**(causal model) is

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + e. \quad (2)$$

e is an unobserved error term that captures everything not contained in $\beta_0 + \beta_1 x_1 + \beta_2 x_2$.

- A1/A2 still hold:

$$E(e) = 0 \quad (\text{A1})$$

$$\text{Cov}(e, x_1) = 0 \quad (\text{A2})$$

$$\text{Cov}(e, x_2) = 0 \quad (\text{A4})$$

Omitted Variables Bias (OVB)

What are the consequences of estimating the model below when x_2 is omitted?

$$y = \beta_0 + \beta_1 x_1 + u. \quad (3)$$

- Because x_2 is not in the model, u now contains x_2 :

$$u = \beta_2 x_2 + e. \quad (4)$$

- We know from earlier analysis that

$$E(\hat{\beta}_1) = \beta_1 + \frac{Cov(u, x_1)}{Var(x_1)}, \quad (5)$$

from which we can write down the bias:

$$\text{Bias}(\hat{\beta}_1) \equiv E(\hat{\beta}_1) - \beta_1 = \frac{Cov(u, x_1)}{Var(x_1)}.$$

Omitted Variables Bias (OVB)

- However, the error term u has changed. Now

$$\begin{aligned}\text{Cov}(u, x_1) &= \text{Cov}(\beta_2 x_2 + e, x_1) \\ &= \beta_2 \text{Cov}(x_2, x_1) + \text{Cov}(e, x_1) \\ &= \beta_2 \text{Cov}(x_2, x_1) + 0.\end{aligned}\tag{6}$$

The last equality uses A2.

- Substituting into (5):

$$\text{Bias}(\hat{\beta}_1) \equiv E(\hat{\beta}_1) - \beta_1 = \beta_2 \frac{\text{Cov}(x_2, x_1)}{\text{Var}(x_1)} = \beta_2 \delta_1,\tag{7}$$

where δ_1 is the “slope” parameter in the population model that relates x_1 and x_2 :

$$x_2 = \delta_0 + \delta_1 x_1 + \text{error}.$$

Omitted Variables Bias (OVB)

$$\text{Bias}(\hat{\beta}_1) \equiv E(\hat{\beta}_1) - \beta_1 = \beta_2 \frac{\text{Cov}(x_2, x_1)}{\text{Var}(x_1)} = \beta_2 \delta_1,$$

If we know *a priori* the signs of β_2 and $\text{Cov}(x_2, x_1)$, we can “sign the bias”:

- If both terms have the same sign (+/+) or (-/-), then the OLS estimator $\hat{\beta}_1$ is **biased upwards**:

$$\text{Bias}(\hat{\beta}_1) > 0.$$

- If both terms have opposite signs (+/-) or (-/+), then the OLS estimator $\hat{\beta}_1$ is **biased downwards**:

$$\text{Bias}(\hat{\beta}_1) < 0.$$

OVB Example

How much should tuition fees be?

In the early 2000s, the claim was that graduates, on average, earn more than £400,000 more than non-graduates over their lifetime. This argument was used by the Government at the time to increase tuition fees to £3,200 per year in 2006. After that, fees increased to £9,000 per year following the October 2010 Spending Review (and are now £9,250 per year).

- £400,000 is only an average and varies by subject, occupation, and other characteristics.
- More generally, what is the return to investment in education (schooling)? 5% per annum? 10% per annum?

OVB Example: The Returns to Education

- Suppose:
 - ▶ y is log of hourly wage (`lhrpay`);
 - ▶ x_1 is years of schooling (`educ`);
 - ▶ x_2 is unobserved innate ability (`abil`).
- Causal model: $\text{lhrpay} = \beta_0 + \beta_1 \text{educ} + \beta_2 \text{abil} + e$
- Estimated model $\text{lhrpay} = \beta_0 + \beta_1 \text{educ} + u$

Does an OLS regression of y (`lhrpay`) on x (`educ`) deliver an unbiased estimate?

OVV Example: The Returns to Education

- $\text{Bias}(\hat{\beta}_1) = \beta_2\delta_1$
- What do we expect?
 - ▶ We expect both variables to have a positive impact on average earnings ($\beta_1 > 0$, $\beta_2 > 0$);
 - ▶ We expect x_1 and x_2 to be positively correlated, as individuals with more innate ability tend to choose/acquire more education ($\delta_1 > 0$).
 - ▶ The estimated effect of an extra year of schooling, on average, is too high:

$$\text{Bias}(\hat{\beta}_1) = \beta_2\delta_1 > 0.$$

- We obtained the following regression results. What can we say about β_1 ?

$$\text{lnhrpay} = 0.584 + 0.083\text{educ}$$

Measurement Error Bias

Occurs everywhere in microeconometrics (indeed, all empirical economics):

- Want the average tax rate but only observe the marginal rate.
- An individual mis-recalling their years of education many years later.
- Age proxying experience in the classic returns-to-education model.

Measurement Error Bias

- We start with the following true model:

$$y = \alpha + \beta x^* + \epsilon. \quad (8)$$

- Exogeneity Assumption:

$$\text{Cov}(\epsilon, x^*) = 0. \quad (\text{A2})$$

We write the " x " variable as x^* to emphasise it is "measured with error":

$$x = x^* + e. \quad (9)$$

- ▶ x^* is the true variable in (8), known as a **latent variable**
- ▶ x is the variable actually in the data
- ▶ e is the **measurement error** (ME)

Measurement Error Bias

- The key assumptions about e are:

$$\text{Cov}(e, x^*) = 0, \quad (\text{A5})$$

$$\text{Cov}(e, \varepsilon) = 0, \quad (\text{A6})$$

and that, on average, individuals report x^* correctly:

$$E(e) = 0.$$

- From (9) and (A5) we get:

$$\text{Var}(x) = \text{Var}(x^*) + \text{Var}(e). \quad (10)$$

Measurement Error Bias

- The estimated model is:

$$y = \alpha + \beta x + u. \quad (11)$$

Using (5) again, we know that

$$\text{Bias}(\hat{\beta}) \equiv E(\hat{\beta}) - \beta = \frac{\text{Cov}(u, x)}{\text{Var}(x)}. \quad (12)$$

- Just as before, we need to work out whether u is correlated with x .
Substituting $x = x^* + e$ into the true model (8):

$$y = \alpha + \beta(x - e) + \varepsilon = \alpha + \beta x + (\varepsilon - \beta e). \quad (13)$$

- Comparing with (11), the term in brackets is the regression error:

$$u = \varepsilon - \beta e. \quad (14)$$

Measurement Error Bias

- Evaluating the numerator in the bias expression (13):

$$\begin{aligned}\text{Cov}(u, x) &= \text{Cov}(\varepsilon - \beta e, x^* + e) \\ &= \text{Cov}(\varepsilon, x^*) + \text{Cov}(\varepsilon, e) - \beta \text{Cov}(e, x^*) - \beta \text{Cov}(e, e) \\ &= 0 + 0 - 0 - \beta \text{Var}(e) \\ &= -\beta \text{Var}(e) \leq 0.\end{aligned}$$

The three zero covariances come from (A2), (A5), and (A6) respectively. $\text{Cov}(u, x)$ can be unambiguously signed, negative if $\beta > 0$ or positive if $\beta < 0$.

- Substituting into (12) and using (10):

$$E(\hat{\beta}) = \beta \left(1 - \frac{\text{Var}(e)}{\text{Var}(x)} \right) = \beta \frac{\text{Var}(x^*)}{\text{Var}(x)} \equiv \beta \lambda, \quad 0 < \lambda < 1. \quad (15)$$

In words, the bias pulls $\hat{\beta}$ towards zero whether or not $\beta > 0$. This is known as **attenuation bias**.

Measurement Error Bias

- Why is λ a positive fraction?

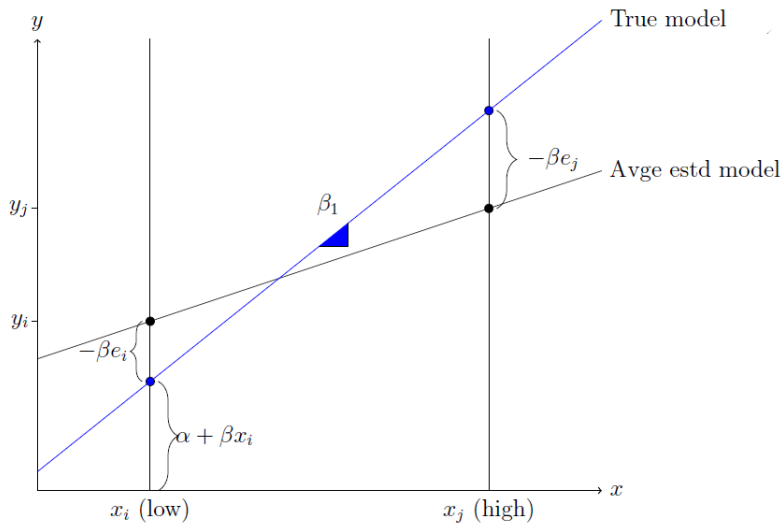
$$\lambda \equiv \frac{\text{Var}(x^*)}{\text{Var}(x^*) + \text{Var}(e)}.$$

- Suppose 20% of the variation in x is due to measurement error. Then $\lambda = 0.8$.
- The correlation/covariance between x and e is key:

$$\begin{aligned}\text{Cov}(e, x) &= \text{Cov}(e, x^* + e) \\ &= \text{Cov}(e, x^*) + \text{Cov}(e, e) \\ &= 0 + \text{Var}(e) > 0.\end{aligned}$$

It is always positive and becomes stronger as the measurement error increases.

Measurement Error Bias



Simultaneity Bias

- This is the idea that x causes y in some economic situations, and at the same time y causes x . In other words, the two variables are determined simultaneously.
- The classic example is a market:

$$y_1 = \alpha_1 y_2 + k_1 + u_1, \quad \alpha_1 < 0, \quad (16)$$

$$y_2 = \alpha_2 y_1 + \beta_2 z_2 + k_2 + u_2, \quad \alpha_2 > 0. \quad (17)$$

where

- ▶ the two equations represent demand and supply respectively;
- ▶ y_1 is quantity and y_2 is price;
- ▶ y_1 and y_2 are endogenous variables (determined in the model);
- ▶ z_2 is exogenous (determined outside the model). In a market for crops, z_2 might be rainfall ($\beta_2 < 0$).

Simultaneity Bias

- There are two endogenous variables, two equations, and two exogeneity assumptions.
- Because z_2 is exogenous:

$$\text{Cov}(u_1, z_2) = 0, \quad (\text{A7})$$

$$\text{Cov}(u_2, z_2) = 0. \quad (\text{A8})$$

- u_1 and u_2 are the unobserved terms in each equation. We assume (for tractability):

$$\text{Cov}(u_1, u_2) = 0.$$

Simultaneity Bias

- The OLS estimator for α_1 is:

$$\hat{\alpha}_1 = \frac{\widehat{\text{Cov}}(y_1, y_2)}{\widehat{\text{Var}}(y_2)}.$$

whose bias is given by the usual formula:

$$\text{Bias}(\hat{\alpha}_1) \equiv E(\hat{\alpha}_1) - \alpha_1 = \frac{\text{Cov}(u_1, y_2)}{\text{Var}(y_2)}.$$

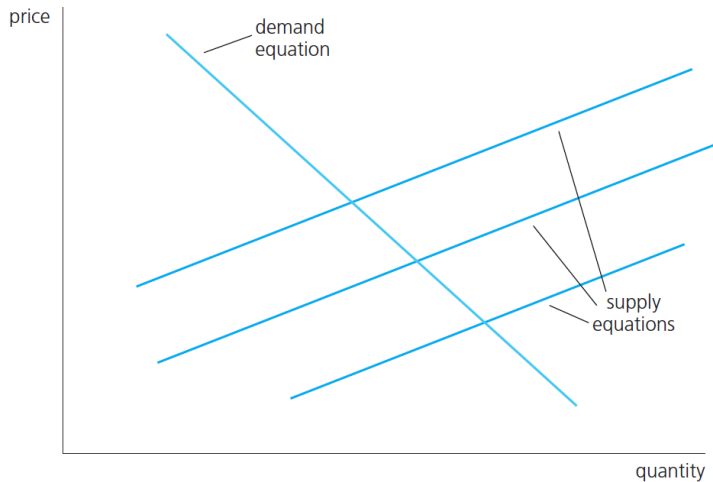
- After a lot of algebra (see the Background Notes),

$$\text{Bias}(\hat{\alpha}_1) = \frac{\alpha_2}{1 - \alpha_1 \alpha_2} \frac{\text{Var}(u_1)}{\text{Var}(y_2)}.$$

- The supply model is upward-sloping, $\alpha_2 > 0$. Typically, $\alpha_1 \alpha_2 < 0$, and so this bias is positive.
- Some intuition is on the next slide ... (Wooldridge's z_1 is our z_2 .)

Simultaneity Bias

FIGURE 16.1 Shifting supply equations trace out the demand equation. Each supply equation is drawn for a different value of the exogenous variable, z_1 .



Simultaneity Bias

- Because y_2 is correlated with u_1 , the OLS estimator suffers from simultaneity bias.
- When $\alpha_1 < 0$ and $\alpha_2 > 0$, the bias is positive.
- Suppose (a) y_1 and y_2 are logged, (b) u_1 explains 80% of the variation in y_2 , and (c) $\alpha_1 = -1$, $\alpha_2 = 1$. Then

$$\text{Bias}(\hat{\alpha}_1) = \frac{\alpha_2}{1 - \alpha_1\alpha_2} \frac{\text{Var}(u_1)}{\text{Var}(y_2)} = \frac{1}{1 - 1 \cdot (-1)} \cdot 0.8 = 0.4.$$

Then

$$E(\hat{\alpha}_1) = \text{Bias}(\hat{\alpha}_1) + \alpha_1 = 0.4 - 1 = -0.6.$$

- OLS estimates the demand curve as more inelastic than it actually is.

Simultaneity Bias

- We need an estimator that exploits the exogenous variation in z_2 , i.e. it traces out the demand curve as z_2 varies.
- (The OLS estimator does not include z_2 in its formula!)
- In the Background Notes, we show that the estimator

$$\tilde{\alpha}_1 = \frac{\text{Cov}(y_1, z_2)}{\text{Cov}(y_2, z_2)}$$

is unbiased:

$$E(\tilde{\alpha}_1) = \alpha_1.$$

- This is **Instrumental Variables**.
- This slide is non-examinable.

Outlook

- There are four main techniques for dealing with endogeneity (discussed above):
 - ▶ **first differences**,
 - ▶ **difference-in-differences**,
 - ▶ **instrumental variables**,
 - ▶ **regression discontinuity**.
- But first, we need to discuss the usefulness of **multiple regression** when modelling dummy variables and categorical variables (Lecture 5).
- And finally, we discuss three of the four techniques (not RD).